# HAIKANG DENG

+1 (919) 260-9698 | frankdenghaikang@gmail.com | Google Scholar | LinkedIn

## EDUCATION

**University of California, Los Angeles** Sep 2024 - Present
Ph.D. in Computer Science, Artificial Intelligence Advisor: Nanyun Peng

**University of North Carolina, Chapel Hill** Aug 2019 - May 2023
B.S. in Computer Science & Statistics, GPA: 3.96/4.0 Advisors: Colin Raffel, Guanting Chen

## PUBLICATIONS

[1] Decoupling Task-Solving and Output Formatting in LLM Generation
**Haikang Deng**, Po-Nien Kung, Nanyun Peng
*ACL*, 2026. Oral Presentation

[2] Learning Structured Reasoning via Tractable Trajectory Control
Po-Nien Kung, Zhen Yang, Jeffrey Luo, Cheng-Fu Yang, **Haikang Deng**, Zi-Yi Dou, Yinfei Yang, Nanyun Peng, Zhe Gan, Kai-Wei Chang
*ICML*, 2026. Spotlight

[3] Lean Graph Search for Formal Theorem Proving
Yihan Xu, Jia-Chen Gu, Junyi Zhang, Di Wu, **Haikang Deng**, Po-Nien Kung, Kai-Wei Chang, Amit Sahai, Wei Wang, Nanyun Peng
*arXiv Preprint, 2026*

[4] From Solvers to Research: Large Language Model-Driven Formal Mathematics at the Research Frontier
Eric Jiang, Xiao Liang, Yingjia Wan, Mengting Li, **Haikang Deng**, et al.
*arXiv Preprint, 2026*

[5] TaoBench: Do Automated Theorem Prover LLMs Generalize Beyond MathLib?
Alexander K. Taylor, Junyi Zhang, Ethan Ji, Vigyan Sahai, **Haikang Deng**, Yuanzhou Chen, Yifan Yuan, Di Wu, Jia-Chen Gu, Kai-Wei Chang, Nanyun Peng, Amit Sahai, Wei Wang
*EMNLP Findings*, 2026

[6] DialectGen: Benchmarking and Improving Dialect Robustness in Multimodal Generation
Yu Zhou, Sohyun An, **Haikang Deng**, Da Yin, Clark Peng, Cho-Jui Hsieh, Kai-Wei Chang, Nanyun Peng
*ResponsibleFM Workshop @ NeurIPS*, 2025

[7] Reward-Augmented Decoding: Efficient Controlled Text Generation With a Unidirectional Reward Model
**Haikang Deng** and Colin Raffel
*EMNLP*, 2023

[8] Large Language Models Struggle to Learn Long-Tail Knowledge
Nikhil Kandpal, **Haikang Deng**, Adam Roberts, Eric Wallace, and Colin Raffel
*ICML*, 2023

## EXPERIENCE

**Bloomberg** Jun 2026 - Sep 2026
*AI/ML Research Intern* New York, NY
· Built routing and specialist agents that traverse decision trees and produce auditable triage outputs.
· Converted technical support runbooks into decision trees to enable deterministic execution.

**University of California, Los Angeles** Sep 2025 - Jun 2026
*Teaching Assistant* Los Angeles, CA
· CS 146: Introduction to Machine Learning (*Fall 2025, Winter 2026*).
· CS 263: Natural Language Processing (*Spring 2026*).

**Amazon** Dec 2023 - Jul 2024
*Software Dev Engineer* Bellevue, WA
· Designed automation for product pre-compute based on given marketplace, merchant, and store utilization score.
· Built a Local Landing Page to display local products available for pick up.

**University of North Carolina, Chapel Hill** Aug 2022 - Mar 2024
*Research Assistant* R-Three Lab
· Conducted research on human feedback alignment, controlled text generation, and retrieval augmentation.
· Trained expert model to route and tag in a contrastive manner with signals from a reward model.